ASSIGNMENT 14.3

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Task 1: AI-Assisted Personal Portfolio Website
Scenario
A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.
Tasks
1. Use GitHub Copilot to generate a semantic HTML structure with sections:
o Home
o About
o Projects
o Contact
2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.
3. Customize AI-generated CSS to add:
o Hover effects on project cards
o Smooth scrolling navigation
Expected Outcome
• Clean and well-structured portfolio website
• Responsive layout for desktop and mobile screens
• Interactive project cards

PROMPT:
<!--Generate a semantic HTML structure with the following sections: Home- About Projects Contact Then, suggest responsive CSS using Flexbox or Grid for layout.customize the CSS to include:Hover effects on project cards (e.g., scale up slightly, add shadow) Smooth scrolling navigation between sections--!>
<!--Ensure the HTML uses semantic tags (header, nav, section, footer) and the CSS is clean, modern, and responsive.--!>

CODE:
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Akhila | Portfolio</title>

<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600Cdisplay=swap"
rel="stylesheet">
```

```
<style>
:root{
  --bg:#0f172a;
  --bg2:#1e293b;
  --text:#ffffff;
  --card:rgba(255,255,255,0.05);
}

.light{
  --bg:#f8fafc;
  --bg2:#e2e8f0;
  --text:#0f172a;
  --card:#ffffff;
}

*{margin:0;padding:0;box-sizing:border-box;font-family:'Poppins',sans-serif;}
body{background:linear-gradient(135deg,var(--bg),var(--bg2));color:var(--text);scroll-
behavior:smooth;transition:0.4s;}


header{display:flex;justify-content:space-between;align-items:center;padding:20px
60px;position:sticky;top:0;background:rgba(255,255,255,0.05);backdrop-filter:blur(10px);z-
index:1000;}

.toggle-btn{
  cursor:pointer;
  font-size:18px;
  color:#ffffff;
  background:none;
  border:none;
  display:inline-flex;
  align-items:center;
  justify-content:center;
  transition:0.3s;
}
.toggle-btn:hover{transform:scale(1.2);}
.toggle-btn:hover{transform:scale(1.1);}

nav a{margin:0 15px;text-decoration:none;color:var(--text);transition:0.3s;}
nav a:hover{color:#38bdf8;}

section{padding:80px 60px;opacity:0;transform:translateY(40px);transition:all 0.8s
ease;position:relative;z-index:1;}
section.show{opacity:1;transform:translateY(0);}
```

```
.hero{text-align:center;padding-top:120px;}
.hero h2{font-size:48px;}
.hero p{margin:10px 0;}

.role{
  color:#38bdf8;
  font-weight:500;
  margin-bottom:15px;
}

.card{background:var(--card);padding:25px;border-radius:15px;margin-top:20px;transition:0.4s;}
.card:hover{transform:translateY(-10px);}

.skills span{display:inline-block;background:#38bdf8;color:#000;padding:8px 12px;border-
radius:20px;margin:5px;font-size:14px;}

.projects{
  display:flex;
  flex-direction:column;
  gap:25px;
}
.project-card{padding:20px;border-radius:15px;background:var(--card);transition:0.4s;}
.project-card:hover{transform:translateY(-12px);}

h2{margin-bottom:20px;}
.skills-clean .skill-item{
  margin:8px 0;
}

.skills-clean .divider{
  height:1px;
  background:rgba(255,255,255,0.1);
  margin:10px 0;
}

.contact-link{
  color:#ffffff;
  text-decoration:none;
  transition:0.3s;
}

.contact-link:hover{
  text-decoration:underline;
  opacity:0.8;
}
```

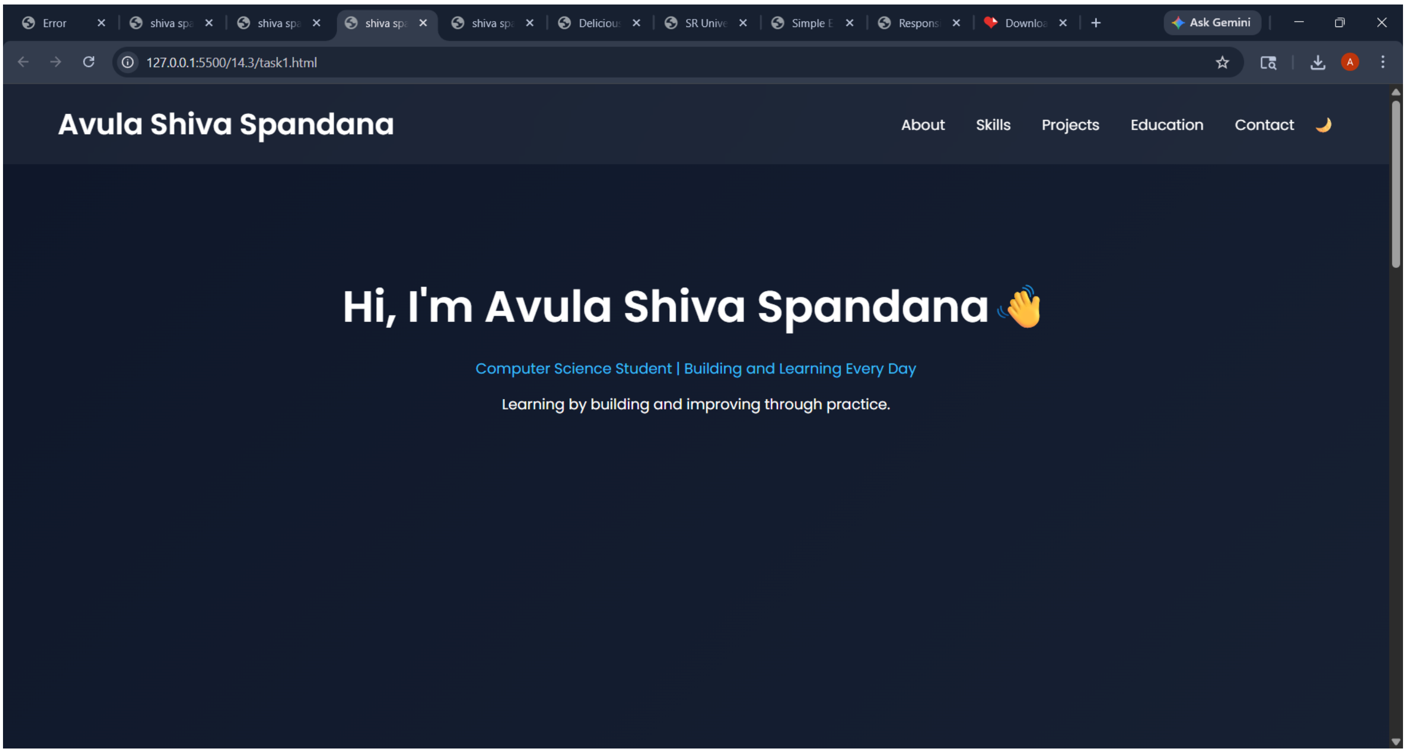
```
footer{text-align:center;padding:20px;color:gray;}
</style>
</head>
<body>


<header>
  <h1>Cherepally Akhila</h1>
  <div>
    <nav style="display:inline-block;">
      <a href="#about">About</a>
      <a href="#skills">Skills</a>
      <a href="#projects">Projects</a>
      <a href="#education">Education</a>
      <a href="#contact">Contact</a>
    </nav>
    <span class="toggle-btn" onclick="toggleTheme()" id="themeIcon"> </span>
  </div>
</header>

<section class="hero show">
  <h2>Hi,  I'm  <span  id="typing"></span>
  <div class="role">Computer Science Student | Building and Learning Every Day</div>
  <p>Learning by building and improving through practice.</p>
</section>

<section id="about">
  <h2>About Me</h2>
  <div class="card">
    <p>
      B.Tech Computer Science student with a strong foundation in Data Structures, Algorithms, and
Machine Learning.
    </p>
    <p>
      Familiar with Python and deep learning frameworks, with hands-on experience building basic
real-time and NLP-based projects.
    </p>
    <p>
      Currently learning and exploring software engineering practices, seeking an internship to gain
practical experience and build scalable solutions.
    </p>
  </div>
</section>
```

```
<section id="skills">
  <h2>Technical Skills</h2>
  <div class="card skills-clean">
    <div class="skill-item"> <strong>Programming:</strong> Python • C • SQL</div>
    <div class="divider"></div>
    <div class="skill-item"> <strong>Core Concepts:</strong> Data Structures • OOP • DBMS</div>
    <div class="divider"></div>
    <div class="skill-item"><strong>Technologies:</strong> Machine Learning • NLP • OpenCV</div>
    <div class="divider"></div>
    <div class="skill-item"> <strong>Frameworks:</strong> PyTorch • TensorFlow</div>
  </div>
</section>

<section id="projects">
  <h2>Projects</h2>
  <div class="projects">

    <div class="project-card">
      <h3>AI vs Real Image Classification</h3>
      <p>
        Developed a deep learning model using MobileNetV2 to classify AI-generated and real-world images.
      </p>
      <p>
        Applied transfer learning with multi-stage fine-tuning and achieved ~87% accuracy.
      </p>
      <p>
        Evaluated using precision, recall, F1-score, and confusion matrix visualization.
      </p>
      <p><strong>Tech:</strong> PyTorch, CNN, Transfer Learning</p>
    </div>

    <div class="project-card">
      <h3>Hybrid Transformer-Capsule Text Classifier</h3>
      <p>
        Built a hybrid NLP model combining Transformer and Capsule Networks for improved text classification.
      </p>
      <p>
        Captured contextual and hierarchical features, outperforming baseline models.
      </p>
      <p>
        Evaluated using accuracy metrics and confusion matrix.
```

```
        </p>
        <p><strong>Tech:</strong> TensorFlow, NLP, Deep Learning</p>
      </div>

      <div class="project-card">
        <h3>Real-Time Activity C Posture Detection</h3>
        <p>
          Developed a real-time system to detect human activities and analyze posture using webcam
input.
        </p>
        <p>
          Used sequence modeling and pose estimation with smoothing techniques for stable
predictions.
        </p>
        <p>
          Implemented posture analysis using joint angle calculations and user calibration.
        </p>
        <p><strong>Tech:</strong> OpenCV, MediaPipe, TensorFlow</p>
      </div>

  </div>
</section>

<section id="education">
  <h2>Education</h2>
   <div class="card">
     <p><strong>SR University</strong> (2023-2027)<br> B.Tech CSE | CGPA: 8.58</p>
     <br>
     <p><strong>Sri Chaitanya Junior College</strong><br>MPC | 953</p>
  </div>
</section>

<section id="contact">
  <h2>Contact</h2>
   <div class="card">
     <p>
        <a href="mailto:cherepallyakhila@gmail.com" target="_blank" class="contact-link">Email</a>
     </p>
     <p>
        <a href="https://www.linkedin.com/in/cherepallyakhila/" target="_blank" class="contact-
link">LinkedIn</a> |
        <a href="https://leetcode.com/u/cherepally/" target="_blank" class="contact-
link">LeetCode</a>
     </p>
  </div>
```

```
</section>

<footer>
  <p>© 2026 Akhila | Portfolio</p>
</footer>

<script>
// Typing Effect
const text = "Akhila";
let i = 0;
function type(){
  if(i < text.length){
    document.getElementById("typing").innerHTML += text.charAt(i);
    i++;
    setTimeout(type,150);
  }
}
type();

// Scroll Reveal
const sections = document.querySelectorAll("section");
window.addEventListener("scroll", () => {
  sections.forEach(sec => {
    const top = sec.getBoundingClientRect().top;
    if(top < window.innerHeight - 100){
      sec.classList.add("show");
    }
  });
});


// Theme Toggle
function toggleTheme(){
  document.body.classList.toggle("light");
  const icon = document.getElementById("themeIcon");
  if(document.body.classList.contains("light")){
    localStorage.setItem("theme","light");
    icon.textContent = "●";
  } else {
    localStorage.setItem("theme","dark");
    icon.textContent = "F";
  }
}

window.onload = function(){
```

```
    const savedTheme = localStorage.getItem("theme");
    const systemDark = window.matchMedia("(prefers-color-scheme: dark)").matches;
    const icon = document.getElementById("themeIcon");

    if(savedTheme === "light" || (!savedTheme CC !systemDark)){
      document.body.classList.add("light");
      icon.textContent = "🟡";
    }
};
</script>

</body>
```




```
</html>
```

OUTPUT:

Task 2: AI-Generated Restaurant Landing Page
Scenario
A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.
Tasks
1. Use Copilot to create a navigation bar with links:
o Home
o Menu
o Gallery

o Contact
2. Generate a menu section styled with CSS cards.
3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.
4. Customize animations or transitions for better user experience.

Expected Outcome

• Fully functional restaurant landing page
• Responsive navigation bar
• Working image slider with smooth transitions

PROMPT:

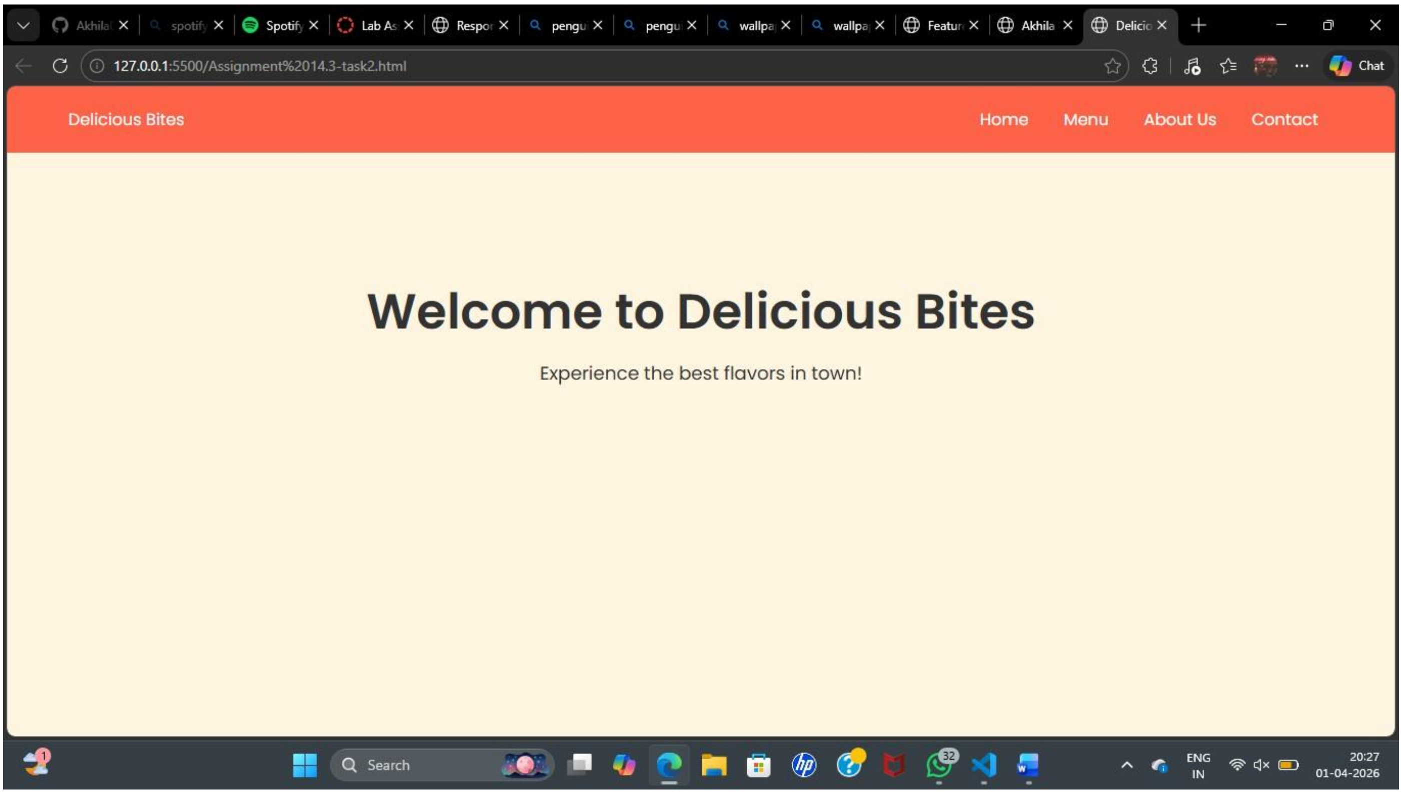
<!-- create a basic HTML5 structure for a restaurant landing page -->

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Delicious Bites | Restaurant</title>
  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600Cdisplay=swap"
rel="stylesheet">
  <style>
    :root {
      --bg: #fff5e1;
      --text: #333;
      --primary: #ff6347;
      --secondary: #ffcccb;
    }
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: 'Poppins', sans-serif;
    }
    body {
      background-color: var(--bg);
      color: var(--text);
    }
    header {
      display: flex;
      justify-content: space-between;
      align-items: center;
      padding: 20px 60px;
      background-color: var(--primary);
```

```
      color: #fff;
    }
    nav a {
      margin: 0 15px;
      text-decoration: none;
      color: #fff;
      transition: 0.3s;
    }
    nav a:hover {
      color: var(--secondary);
    }
    .hero {
      text-align: center;
      padding-top: 120px;
    }
    .hero h1 {
      font-size: 48px;
    }
    .hero p {
      margin-top: 10px;
      font-size: 18px;
    }
  </style>
</head>
<body>

  <header>
    <div class="logo">Delicious Bites</div>
    <nav>
      <a href="#home">Home</a>
      <a href="#menu">Menu</a>
      <a href="#about">About Us</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <section class="hero" id="home">
    <h1>Welcome to Delicious Bites</h1>
    <p>Experience the best flavors in town!</p>
  </section>

  <!-- Additional sections for menu, about us, and contact can be added here -->
</body>
</html>
```

OUTPUT:



Task 3: AI-Powered Event Registration Form
Scenario
SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.
Tasks
1. Ask Copilot to generate an HTML form with the following fields:
o Name
o Email
o Phone Number
o Department
o Event Selection
2. Use AI assistance to style the form using CSS for readability and accessibility.
3. Implement JavaScript validation using Copilot:
o Email format validation
o Phone number length validation
o Required field checks
Expected Outcome
• User-friendly and visually appealing form
• Real-time validation messages
• Prevention of invalid submissions'

PROMPT:
<!--Create an online registration form for SR University's technical fest with fields for Name, Email,

Phone Number, Department, and Event Selection; style the form using CSS for readability and accessibility; implement JavaScript validation for email format, phone number length, and required field checks; ensure real-time validation messages and prevention of invalid submissions so the form is user-friendly, visually appealing, and fully functional.
-->

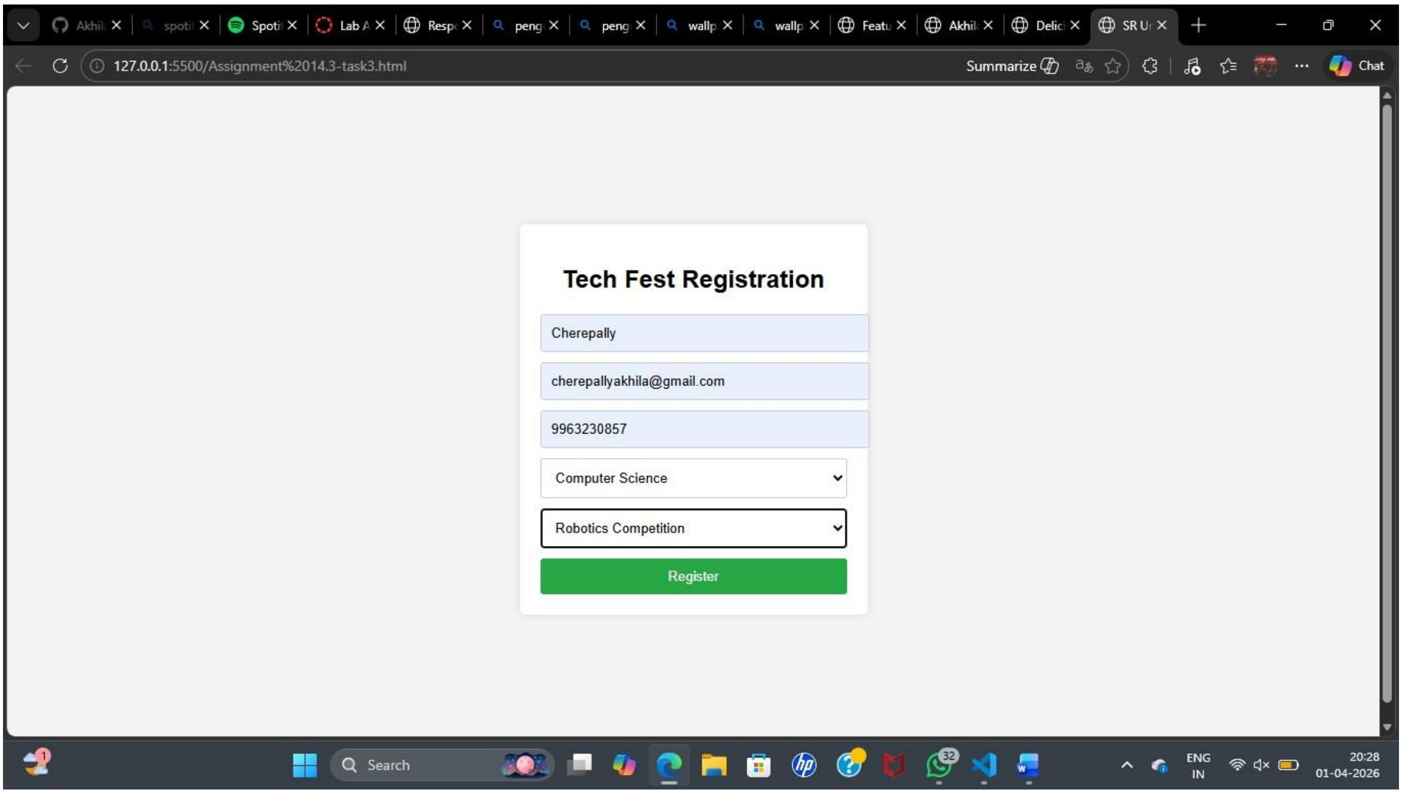
CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SR University Tech Fest Registration</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }
    .registration-form {
      background-color: #fff;
      padding: 20px;
      border-radius: 5px;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
      width: 300px;
    }
    .registration-form h2 {
      text-align: center;
      margin-bottom: 20px;
    }
    .registration-form input, .registration-form select {
      width: 100%;
      padding: 10px;
      margin-bottom: 10px;
      border: 1px solid #ccc;
      border-radius: 3px;
    }
    .registration-form button {
      width: 100%;
      padding: 10px;
      background-color: #28a745;
      color: #fff;
```

```
      border: none;
      border-radius: 3px;
      cursor: pointer;
    }
    .registration-form button:hover {
      background-color: #218838;
    }
    .error-message {
      color: red;
      font-size: 12px;
    }
  </style>
</head>
<body>
  <form class="registration-form" id="regForm">
    <h2>Tech Fest Registration</h2>
    <input type="text" id="name" placeholder="Name" required>
    <div class="error-message" id="nameError"></div>

    <input type="email" id="email" placeholder="Email" required>
    <div class="error-message" id="emailError"></div>

    <input type="text" id="phone" placeholder="Phone Number" required>
    <div class="error-message" id="phoneError"></div>

    <select id="department" required>
      <option value="">Select Department</option>
      <option value="CSE">Computer Science</option>
      <option value="ECE">Electronics</option>
      <option value="ME">Mechanical</option>
      <!-- Add more departments as needed -->
    </select>
    <div class="error-message" id="deptError"></div>

    <select id="event" required>
      <option value="">Select Event</option>
      <option value="coding">Coding Challenge</option>
      <option value="robotics">Robotics Competition</option>
      <option value="hackathon">Hackathon</option>
      <!-- Add more events as needed -->
    </select>
    <div class="error-message" id="eventError"></div>

    <button type="submit">Register</button>
  </form>
```

```
    <script src="validation.js"></script>
</body>
</html>
```

OUTPUT:



Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an "Add to Cart" feature using JavaScript.
3. Modify AI-generated code to include:

o Cart item counter at the top-right corner

o Visual feedback when a product is added

Expected Outcome

• Functional product catalog

• Working cart counter

• Interactive user experience

PROMPT: <!--Create a basic e-commerce product page with a grid-based product catalog using HTML and CSS, implement an "Add to Cart" feature with JavaScript, include a cart item counter at the top-right corner, and provide visual feedback when a product is added; ensure the catalog is functional, the cart counter works correctly, and the overall user experience is interactive and responsive.

-->

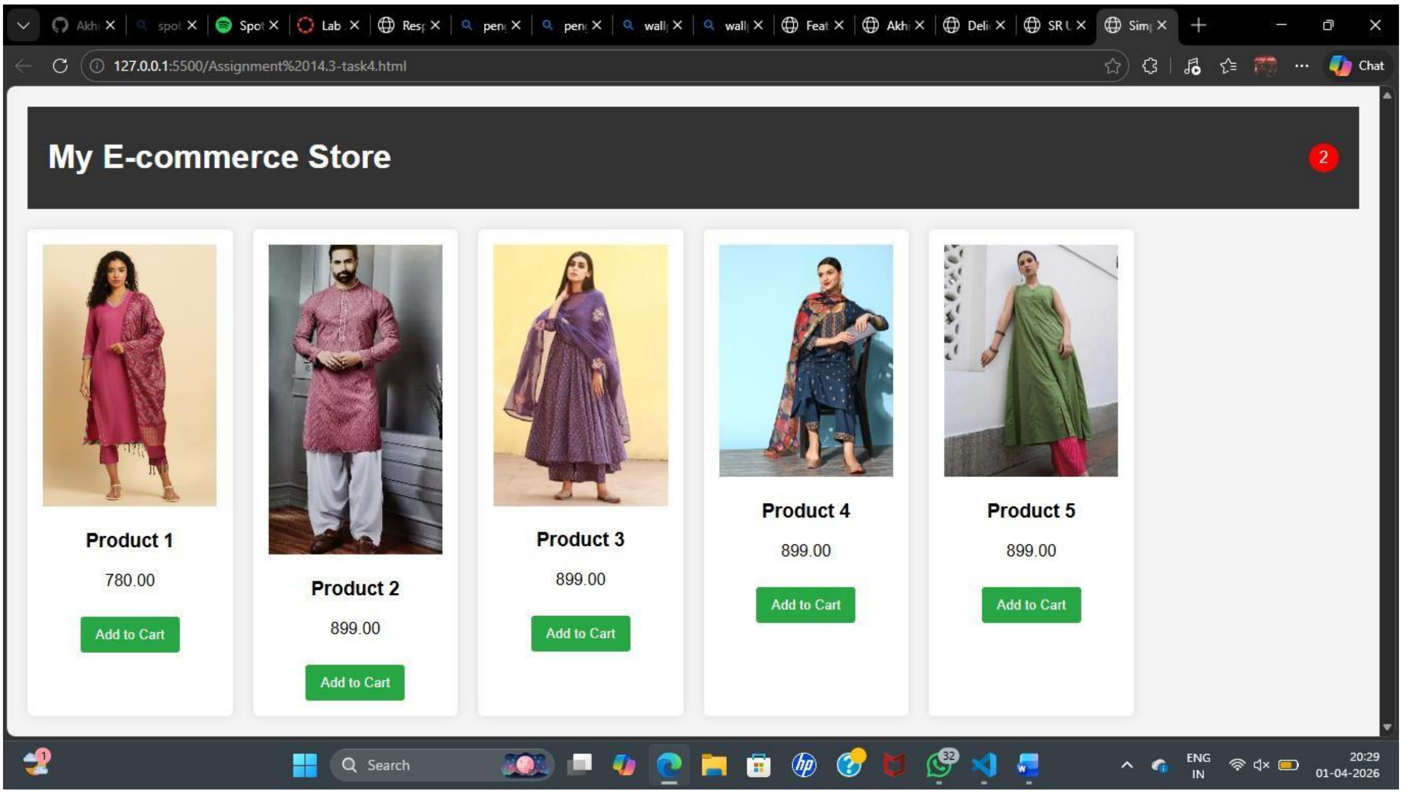
CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Simple E-commerce Product Page</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      margin: 0;
      padding: 20px;
    }
    .header {
      display: flex;
      justify-content: space-between;
      align-items: center;
      background-color: #333;
      color: #fff;
      padding: 10px 20px;
    }
    .cart-counter {
      background-color: red;
      border-radius: 50%;
      padding: 5px 10px;
      color: #fff;
    }
    .product-grid {
      display: grid;
      grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
      gap: 20px;
      margin-top: 20px;
    }
    .product-card {
      background-color: #fff;
      padding: 15px;
      border-radius: 5px;
      box-shadow: 0 0 10px rgba(0,0,0,0.1);
      text-align: center;
    }
    .product-card img {
```

```
        max-width: 100%;
        height: auto;
      }
      .add-to-cart {
        margin-top: 10px;
        padding: 10px 15px;
        background-color: #28a745;
        color: #fff;
        border: none;
        border-radius: 3px;
        cursor: pointer;
      }
      .add-to-cart:hover {
        background-color: #218838;
      }
    </style>
</head>
<body>
    <div class="header">
      <h1>My E-commerce Store</h1>
      <div class="cart-counter" id="cartCounter">0</div>
    </div>

    <div class="product-grid">
      <div class="product-card">
        <img src="https://wforwoman.com/cdn/shop/files/23AUWS19527-
220351_2.jpg?v=1721307670" alt="Product 1">
        <h3>Product 1</h3>
        <p>780.00</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
      </div>
      <div class="product-card">
        <img
src="https://th.bing.com/th/id/OIP.mjCY7ccDSDVdlvhLCMJU1QHaNK?w=182Ch=324Cc=7Cr=0Co=7
Cpid=1.7Crm=3" alt="Product 2">
        <h3>Product 2</h3>
        <p>899.00</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
      </div>
      <!-- Add more products as needed i want to add 3 more pics-->
      <div class="product-card">
        <img
src="https://tse1.mm.bing.net/th/id/OIP.yEVSK8D_yB8EKSR6IMVZ7QHaLH?rs=1Cpid=ImgDetMainCo
=7Crm=3" alt="Product 3">
        <h3>Product 3</h3>
```

```
        <p>899.00</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
      </div>
      <div class="product-card">
        <img
src="https://assets.myntassets.com/h_200,w_200,c_fill,g_auto/h_1440,q_100,w_1080/v1/assets/ima
ges/20148332/2022/10/13/7e66831f-6330-4a88-8b13-0b1d8c19802216656416 81476-Varanga-
Women-Blue-Embroidered-Kurta-with-Trousers--With-Dup-1.jpg" alt="Product 4">
        <h3>Product 4</h3>
        <p>899.00</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
      </div>
      <div class="product-card">
        <img
src="https://tse4.mm.bing.net/th/id/OIP.If7VQXeEJCYyNmXIfALoGgHaJ4?rs=1Cpid=ImgDetMainCo=7
Crm=3" alt="Product 5">
        <h3>Product 5</h3>
        <p>899.00</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
      </div>
    </div>

    <script>
      let cartCount = 0;

      function addToCart() {
        cartCount++;
        document.getElementById('cartCounter').innerText = cartCount;
        alert('Product added to cart!');
      }
    </script>
</body>
</html>
```

OUTPUT:



Task 5: Create a Responsive Web Page Layout

Instructions:

- Design a basic web page layout with a header, main content area, and footer using HTML and CSS.
- Use AI to assist in generating responsive CSS for different screen sizes.
- Ensure the layout is clean and visually organized.

Expected Output:

- A responsive web page with:
- Header with navigation links
- Main section with placeholder text/images
- Footer with copyright or contact info
- Layout adapts correctly to desktop, tablet, and mobile screen sizes.

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments C code explanation, and output and if required, screenshots.

PROMPT:

<!--Design a basic web page layout with a header containing navigation links, a main content area with placeholder text and images, and a footer with copyright or contact info; use HTML and CSS with responsive styling for desktop, tablet, and mobile screen sizes; ensure the layout is clean, visually organized, and adapts correctly across devices with smooth readability and accessibility. English
-->

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Responsive Web Page Layout</title>
<style>
  body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 0;
    background-color: #f4f4f4;
  }
  header {
    background-color: #333;
    color: #fff;
    padding: 20px;
    text-align: center;
  }
  nav a {
    color: #fff;
    margin: 0 15px;
    text-decoration: none;
  }
  main {
    padding: 20px;
  }
  .placeholder {
    background-color: #ccc;
    height: 200px;
    margin-bottom: 20px;
  }
  footer {
    background-color: #333;
    color: #fff;
    text-align: center;
    padding: 10px;
    position: fixed;
    width: 100%;
    bottom: 0;
  }
  @media (max-width: 768px) {
    nav a {
      display: block;
      margin: 10px 0;
    }
  }
</style>
```

```
</head>
<body>
  <header>
    <h1>ANUV JAIN</h1>
    <nav>
      <a href="#">Home</a>
      <a href="#">About</a>
      <a href="#">Songs</a>
      <a href="#">Contact</a>
    </nav>
  </header>
  <main>
    <div class="image-placeholder">
      <img src="https://images.travelandleisureasia.com/wp-
content/uploads/sites/2/2023/01/16162848/Featured-Inside-1-15-1600x900.jpeg" alt="Image
Placeholder 1">
    </div>
  </main>
  <footer>
    Ccopy; 2024 My Responsive Web Page. All rights reserved.
  </footer>

</html>
```

OUTPUT: